

State Space Analysis & Routh-Hurwitz Stability

Subject: Control Systems Engineering (EC3502) | Unit 1

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CORE CONCEPTS & LECTURE SUMMARY:

- Open loop vs closed loop transfer function derivation using block reduction.
- Mason's Gain formula with forward paths and non-touching feedback loops.
- Routh-Hurwitz algebraic array for absolute stability determination.
- State vector differential equation: $\dot{x} = Ax + Bu$; $y = Cx + Du$.